14 A student does an experiment to determine the specific heat capacity of a metal block.

She records the following measurements.

energy supplied = 12 kJ

initial temperature = 20 °C

final temperature = 45 °C

mass of metal block = 600 g

What is the specific heat capacity of the metal block?

A 8.0×10^{-4} J / (kg °C)

B 0.80 J / (kg °C)

C 180 J / (kg °C)

D 800 J / (kg °C)